

Concrete is 8% of global CO₂. Most of the cement in it is *insurance*.

Half of cement's emissions come from the chemical reaction itself, so renewables cannot abate them. Industrial wastes can replace much of the cement, but site engineers over-dose it anyway, because predicting concrete strength is hard and being wrong is unthinkable. CarbonCast learns that relationship from 1,030 real lab mixes and finds the lowest-carbon mix that still clears the target at the **calibrated 5th percentile, not the average**.

54%

LESS CO₂ AT IDENTICAL M30 STRENGTH

977 t

CO₂E SAVED ON ONE VIT ACADEMIC BLOCK

THE PROBLEM

Invisible insurance, poured into everything

Cement is ~8% of all human CO₂. If it were a country it would rank third. Fly ash and slag, waste from coal plants and steel furnaces, can replace much of it at near-zero carbon cost.

They are under-used because strength prediction is hard, so engineers add cement as a safety buffer. That buffer is never measured. It is just poured.

THE METHOD

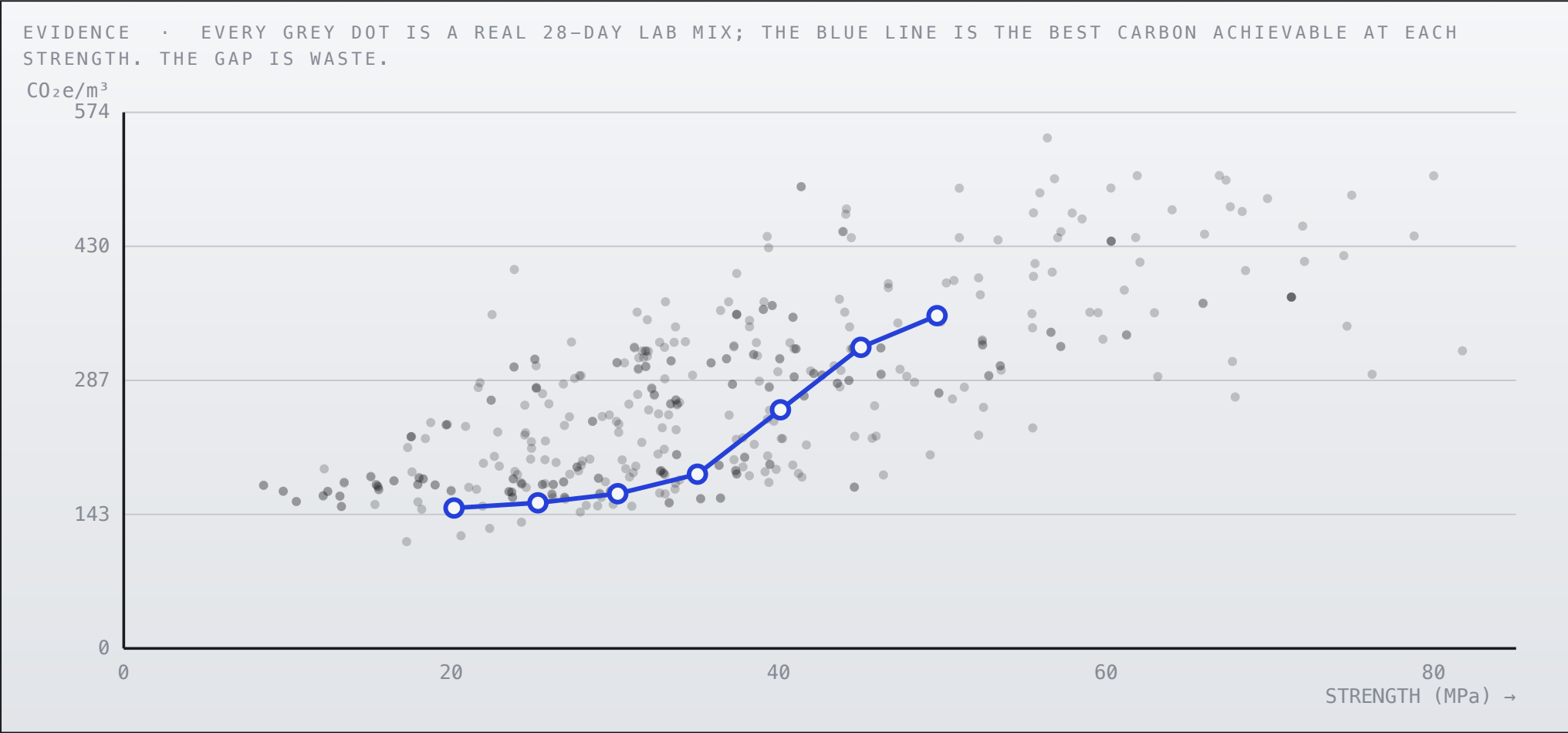
Optimise the margin, don't assume it

1. Learn strength from mix on **1,030** real lab results

2. **Calibrate** the uncertainty so the confidence band is honest

3. Search for minimum carbon, constrained on the **lower bound**

4. **Refuse** any mix the training data cannot vouch for



RESULTS

GRADE	TYPICAL	CARBONCAST	CUT	WASTE USED
M20	301	150	50%	58%
M25	319	156	51%	65%
M30	359	166	54%	60%
M40	409	255	38%	47%

kg CO₂e per m³. M30 is also **₹1,227/m³ cheaper**. The typical M40 mix has a 5th-percentile strength of just 38.6 MPa: it does not reliably make its own grade. Ours clears 40.1 with 38% less carbon.

THE MODEL

R² on 206 unseen mixes

0.9228

RMSE

±4.46 MPa

Raw interval coverage

69%

After conformal calibration

87% (target 90%)

Gradient-boosted trees. Raw quantile models were overconfident, which would have made our safety margin fiction. Split-conformal quantile regression widened the band by 2.97 MPa for a distribution-free guarantee.

